

NORFOLK INTL, VA, USA

WMO: 723080

Lat: 36.903N

Lon: 76.192W

Elev: 9

StdP: 101.22

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13737

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.6	-3.5	-16.4	0.9	-3.2	-13.5	1.2	-0.8	12.1	11.0	11.0	10.8	4.1	350	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	34.2	24.9	32.9	24.5	31.7	24.0	26.3	31.3	25.7	30.5	25.2	29.8	4.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.1	28.4	24.4	19.3	27.9	23.9	18.7	27.5	81.4	31.2	78.9	30.6	76.6	29.9	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.8	9.2	8.3	DB	-8.4	36.8	2.8	1.3	-10.4	37.7	-12.0	38.5	-13.6	39.2	-15.6	40.1	
			WB	-10.0	27.4	2.5	0.8	-11.8	28.0	-13.2	28.5	-14.7	28.9	-16.5	29.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	5.3	6.6	10.1	15.4	20.0	24.6	26.7	26.0	23.1	17.5	11.6	7.8
	DBStd	8.65	5.72	5.23	5.19	4.75	4.08	3.18	2.40	2.25	2.79	4.10	4.54	5.07
	HDD10.0	500	169	121	65	7	0	0	0	0	1	33	104	
	HDD18.3	1744	405	329	261	113	31	2	0	2	66	208	327	
	CDD10.0	2787	23	26	67	171	308	438	519	495	392	233	80	37
	CDD18.3	991	1	1	5	27	82	189	261	237	143	39	4	2
	CDH23.3	8287	1	7	41	222	619	1653	2588	2075	853	204	18	6
	CDH26.7	2854	0	0	4	57	192	616	1035	702	210	38	1	0
(14) Wind	WSAvg	4.0	4.2	4.3	4.6	4.7	4.2	3.8	3.6	3.4	3.9	3.9	3.9	3.9
(15) Precipitation	PrecAvg	1153	94	81	94	79	93	100	137	131	105	85	71	80
	PrecMax	1650	252	208	263	184	257	247	365	364	350	257	145	158
	PrecMin	673	27	16	13	11	16	28	9	19	7	0	2	17
	PrecStd	210	41	38	48	41	43	52	79	82	80	52	39	36
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	24.3	27.0	30.8	32.8	35.2	36.4	35.0	32.8	29.7	25.5	24.0
		MCWB	17.4	17.8	17.9	19.9	22.7	24.6	25.5	25.5	23.8	22.2	19.1	18.4
	2%	DB	19.9	20.8	24.2	28.3	30.8	33.4	34.5	33.2	30.8	27.7	23.0	20.8
		MCWB	16.2	14.9	16.4	19.3	22.0	24.1	25.3	25.0	23.6	21.7	18.2	17.3
	5%	DB	17.4	18.2	21.5	26.0	28.9	31.9	33.1	31.9	29.0	25.6	20.9	18.6
		MCWB	14.5	14.0	15.4	18.3	21.4	23.7	24.9	24.5	23.0	20.7	17.1	16.0
	10%	DB	14.7	15.5	18.9	23.4	26.9	30.2	31.7	30.4	27.5	23.8	18.9	16.2
		MCWB	12.0	11.6	14.3	17.3	20.3	23.2	24.6	24.1	22.5	19.7	15.7	13.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	18.8	19.6	22.2	24.7	26.3	27.3	27.0	25.6	24.6	21.1	19.9
		MCDB	20.9	22.0	23.9	27.5	29.7	31.0	32.8	31.8	29.3	27.5	23.2	22.4
	2%	WB	16.8	16.8	18.1	20.6	23.3	25.4	26.5	26.1	24.9	23.1	19.4	18.0
		MCDB	19.2	20.0	22.1	25.8	28.1	30.7	31.9	30.7	28.2	25.7	21.8	20.2
	5%	WB	14.8	14.6	16.7	19.5	22.3	24.7	25.8	25.5	24.3	21.9	17.8	16.4
		MCDB	17.2	17.5	20.6	24.3	27.0	29.8	31.0	29.9	27.3	24.6	20.3	18.3
	10%	WB	12.1	11.8	14.8	18.2	21.4	24.1	25.3	25.0	23.7	20.5	16.1	14.0
		MCDB	14.5	14.7	18.4	22.6	25.8	28.7	30.3	29.0	26.4	23.1	18.5	16.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	8.8	9.1	9.7	8.8	8.2	7.9	7.2	6.8	8.2	8.5	8.2
		MCDBR	12.7	13.8	13.8	13.2	11.6	10.7	10.0	9.4	9.0	10.1	11.4	11.2
	5% WB	MCWBR	9.8	9.6	8.2	6.3	5.1	4.0	3.5	3.5	3.9	5.0	7.1	8.7
		MCWBR	11.7	12.6	12.0	11.5	10.0	9.1	8.8	7.9	7.1	7.9	9.3	10.6
(40) Clear-Sky Solar Irradiance	taub	taub	0.309	0.321	0.348	0.391	0.434	0.482	0.510	0.503	0.407	0.356	0.325	0.314
		taud	2.541	2.503	2.460	2.351	2.260	2.177	2.115	2.119	2.380	2.515	2.551	2.559
	Ebn at Noon	Ebn at Noon	888	917	919	891	853	809	783	779	845	865	861	858
		Edh at Noon	78	91	104	122	136	148	156	153	112	89	77	72
(44) All-Sky Solar Radiation	RadAvg	2.34	3.10	4.15	5.29	5.96	6.39	6.09	5.49	4.48	3.55	2.64	2.08	
	RadStd	0.19	0.33	0.33	0.38	0.51	0.41	0.29	0.32	0.36	0.40	0.21	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	+168	+108
(47) Regional (54 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	+166	+105

Nomenclature: See separate page